EXPLORATORY CASE STUDY · COST AND MANAGEMENT ACCOUNTING

Should Milky Mist start its own dairy farm?

A 15-acre pilot farm in Tamil Nadu: what it costs, what it saves, and what has to go right

Harshavardhaan | Qualified CMA

November 2026

The answer in brief

A pilot of 1,400 HF crossbred cows on 15 acres gives about 23,800 litres of milk a day: under 1% of Milky Mist's milk, but about 6% of what its Skyr and Greek yoghurt need. Farm milk costs about a third less than bought milk, and the farm pays back in about 7 years, but after the cost of capital it roughly breaks even. It needs more than 20 litres a cow, or slightly cheaper feed, to add value; its real case is learning and traceable premium milk.

₹57 cr	₹38.9 / L	−₹1.5 cr	19.1%	20.2 L
capital, FY28–FY30	farm milk cost, FY31 (bought: ₹57.6)	NPV at 19.72% pre-tax WACC	IRR	break-even yield per cow

About this study

This is an independent exploration based on publicly available information (annual report, earnings calls, IPO documents, news and public price listings) and my own stated assumptions. It uses no company data, is not affiliated with or endorsed by Milky Mist Dairy Food Limited, and is not investment advice. All figures are estimates that would change with real quotes and site data.

Use of AI

I used Claude (Anthropic) as a working tool to speed up research, calculations, model building and drafting. I set the questions, made the decisions, checked the numbers, and worked through every process in the model until I understood how and why it works. The full split of work is in the AI and human contribution report.

Project Report

Milky Mist: a 15-acre pilot dairy farm | Harshavardhaan, Qualified CMA | November 2026

Background

Milky Mist is one of India's fastest-growing dairy brands, with products across the traditional and emerging dairy value chain such as paneer, cheese, curd, ghee, butter and high-protein yoghurts. It mainly sells value-added products, and its primary raw material is milk. In FY26 it procured about 39.6 crore litres, up about 29% on the year before, from more than 92,000 farmers and through contract farming. With management expecting strong growth, its milk need keeps rising.

Problem statement

Does a small own dairy farm make sense for Milky Mist, and what does it give the business after all the resources put in?

An own farm cannot replace a meaningful share of the company's milk: it would need hundreds of acres and thousands of cows. Its plants also process milk from all sources together, so a farm cannot be tied to one plant. So this project sizes the farm as a **pilot on 15 acres**, within Tamil Nadu's land ceiling for companies, and shows its milk against the requirement of individual products such as Skyr / Greek yoghurt and paneer.

The farm is judged on **savings as part of the business**: the cost of the milk the company no longer buys, less everything the farm costs to build and run, with inflation included. The goal is not to prove the farm is profitable, but to see what it really costs and what has to go right.

Walkthrough

Sales, volume and milk requirement

We forecast sales and volume for each product from FY26 using management's commentary, easing the growth over time because no company keeps growing 30–40% a year for a decade. Milk per kilogram of each product turns volumes into milk. Total milk need grows from about 10.9 lakh litres a day in FY26 to 28.8 lakh in FY31 and 58.6 lakh in FY40. Skyr / Greek yoghurt, the fastest-growing product, needs about 4 lakh litres a day by FY31.

Base case	FY26	FY27F	FY31	FY35	FY40
Sales (₹ cr)	3,137	4,288	9,850	16,508	27,958
Total milk need (lakh L a day)	10.9	14.3	28.8	41.8	58.6
Paneer milk (lakh L a day)	4.04	5.18	9.68	13.92	19.52
Skyr / Greek milk (lakh L a day)	0.35	0.99	4.02	6.44	9.46

Why 15 acres

Two reasons. First, **15 standard acres is the most a company can hold in Tamil Nadu without government permission under section 37-A** of the land ceiling law, so the pilot needs no land-ceiling approval. Second, 15 acres fits about 1,250–1,500 cows in compact free-stall barns with all fodder bought in, enough to matter for a product line. We settled on **1,400 cows**, giving about **23,800 litres a day** from FY31: about 0.8% of total milk, 2.5% of paneer milk and 5.9% of Skyr / Greek milk in FY31.

Farm milk as a share of …	FY31	FY35	FY40
Total milk need	0.8%	0.6%	0.4%
Skyr / Greek yoghurt milk	5.9%	3.7%	2.5%
Paneer milk	2.5%	1.7%	1.2%

Timeline

- **FY28:** approvals, land, most of the construction and the first batch of about 467 cows.
- **FY29:** second batch; the first batch starts milking.
- **FY30:** third batch; the first two batches milk.
- **FY31 onwards:** the full herd of 1,400 milking; the model runs to FY40.

Land and approvals

About 15 acres of dry land at Gobichettipalayam, 32 km from the Perundurai plant, at about ₹45 lakh an acre plus 9% stamp duty and registration. Because the farm stays within 15 standard acres, no section 37-A land-ceiling permission is needed. Approvals still needed: TNPCB consent (a farm with more than 500 animals is Red category), groundwater permission and local building approvals.

Cow herd

All cows are **HF crossbred (62.5–75% Holstein)**, chosen for heat tolerance and high yield, bought from breeders in Tamil Nadu, Karnataka and Kerala at about ₹90,000 each. 85% of adult cows are in milk at any time, averaging **20 litres a day** over the lactation. From FY31 about 25% of the herd is replaced each year by heifers raised on the farm, so about 926 young stock are kept, and culls, surplus heifers and male calves are sold.

Head (annual average)	FY28	FY29	FY30	FY31	FY33 on
Adult cows	467	933	1,400	1,400	1,400
Cows in milk	–	397	793	1,190	1,190
Young stock	–	139	371	649	926
Milk (litres a day)	–	7,140	15,070	23,800	23,800

Feed

Feed is about three-quarters of the running cost. A milking cow's ration costs about ₹405 a day at FY27 prices (silage, concentrate, straw and minerals), sized for 20 litres; above that each extra litre needs about half a kilogram more concentrate.

FY31 cost per litre	₹ per litre
Feed	30.99
Staff	3.11
Milk transport	1.53
Vet, AI, power, water, insurance, repairs, other	5.02
Less: by-products (compost, calves, culls)	(1.73)
Net farm cost	**38.91**
Bought milk (landed)	57.57
Saving per litre	**18.65**

Water, power and waste

Borewells supply about 200 KL a day, with canal water and tankers as backup, and a treatment and reuse loop keeps all wastewater on the farm. 500 kW of rooftop solar covers the daytime load. Manure is composted and sold.

Capital

The farm costs **₹52.7 cr at FY27 prices** (₹57.1 cr as spent over FY28–FY30): land ₹7.4 cr, buildings ₹20.5 cr, equipment and solar ₹7.6 cr, cows ₹13.1 cr, and pre-operative costs and contingency ₹4.1 cr. That is about ₹3.76 lakh per cow. It is paid from company cash.

Capital	FY27 prices (₹ cr)	As spent (₹ cr)
Land, 15 acres	7.43	7.88
Buildings and site works	20.49	22.11
Equipment and 500 kW solar	7.58	8.24
1,400 cows	13.09	14.44
Pre-operative and contingency	4.07	4.39
Total	**52.66**	**57.07**

Costing method

- Savings = milk purchases avoided, less farm running costs, plus by-products, less capital; land and herd counted back at market value at the end of FY40.
- Each cost and the bought milk price inflate at their own historical average (milk 5.2%, feed 6%, labour 6%, power 3.5%, others 5%).
- Nominal, pre-tax cash flows, discounted to the start of FY28 at a pre-tax WACC of 19.72%: a post-tax WACC of 14.75% on market values, using my own 5-year beta regression on Parag Milk Foods relevered to Milky Mist and Blume-adjusted, grossed up for the 25.17% tax rate so the rate matches the cash flows.
- Bought milk is valued at ₹47 a litre landed at FY27 prices, and farm milk is taken to be of the same quality.
- Scenarios move several assumptions together; sensitivities test yield and feed cost, the two biggest levers.

Outcomes

Measure	Upside	Base	Downside
Farm milk cost per litre, FY31	₹34.91	**₹38.91**	₹50.96
Bought milk price, FY31	₹62.51	**₹57.57**	₹53.97
NPV at 19.72% pre-tax WACC	₹42.1 cr	**−₹1.5 cr**	−₹63.6 cr
IRR	35.4%	**19.1%**	none

₹ cr	FY28	FY29	FY30	FY31	FY35	FY40
Milk purchases avoided	–	13.6	30.1	50.0	61.3	78.9
Farm costs less by-products	(4.9)	(14.8)	(26.7)	(33.8)	(43.8)	(58.2)
Capital	(31.9)	(14.1)	(11.1)	–	–	–
Working capital, land and herd value	(0.4)	(0.8)	(1.0)	(0.6)	(0.2)	42.9
Net cash flow	**(37.2)**	**(16.2)**	**(8.6)**	**15.6**	**17.3**	**63.6**

NPV (₹ cr) by yield and feed cost, everything else at base:

Yield ↓ / feed cost →	90%	95%	100%	105%	110%
18 L	−8.2	−13.4	−18.7	−24.0	−29.3
19 L	0.5	−4.8	−10.1	−15.4	−20.7
20 L	9.1	3.8	**−1.5**	−6.8	−12.0
21 L	15.4	9.9	4.5	−0.9	−6.3
22 L	21.6	16.1	10.5	5.0	−0.6

- **Roughly break-even:** NPV −₹1.5 cr, with an IRR of 19.1% against a 19.72% pre-tax hurdle; payback in about 6.9 years from the start of FY28, and about ₹158 cr of net cash saved by FY40 before discounting.
- **Cheaper milk:** about ₹18.7 a litre below bought milk from FY31, or about ₹16 cr a year.
- **Two levers decide it:** NPV breaks even at 20.2 litres a cow, or with feed 1.4% below plan.
- **A small part of our milk:** under 1% of total milk, but about 6% of Skyr / Greek yoghurt milk. Farmers and contract farms remain the main source.
- **Recommendation:** take the pilot to a feasibility stage only; it is justified more by learning and traceable premium milk than by its return. Commit only after land, a fixed-price feed contract and proven 20-litre cows are confirmed, and check batch 1's yields before buying batch 3.

Benchmark: Parag Milk Foods' own farm

Parag Milk Foods is the closest listed peer that runs its own dairy farm, and its experience supports the main choices here.

- **Scale in steps.** Parag grew its Bhagyalakshmi Dairy Farm near Pune from about 1,000 cows in 2005 to over 5,000 by 2025. A 1,400-cow pilot bought in three batches is close to where Parag started.
- **Water comes first.** Parag sited its farm near the Bheema river and is building a rainwater reservoir at its new farm. Our plan relies on borewells, canal backup, a reuse loop and rainwater harvesting from day one.
- **The yield is in line.** Parag's target of 2.5 lakh litres a day from 20,000 cows works out to about 20 litres per cow in milk, the same as our base case.
- **The farm pays through a premium brand.** Parag sells its farm milk as Pride of Cows, delivered to homes at about ₹120–140 a litre. If our farm milk went into a premium, traceable line, it would be worth far more than the ₹47 bought-milk value the model uses. That upside is not counted here.

Skills shown

Skill	Where it shows in this project
Cost accounting	Cost per litre built up by element (feed, staff, power, transport and more), by-product credits, ration-linked feed cost, make-or-buy against bought milk
Capital budgeting	Phased capital, savings-based cash flows, NPV, IRR, payback, discounted payback, residual values, working capital
Cost of capital	Beta regression in Python against Parag Milk Foods, unlevering and relevering at market D/E, Blume adjustment, CAPM, market-value WACC, pre-tax gross-up, four-peer cross-check
Forecasting	Product-wise sales, volume and milk requirement to FY40 from management commentary, eased to realistic growth
Financial modelling	Linked nine-sheet Excel model with inputs, scenarios, sensitivities and an independent accuracy check
Risk analysis	Upside and downside scenarios, a yield and feed cost grid, and break-evens for yield, feed and milk price
Business judgement	Sizing the farm to what the company can manage, reframing the question as evidence came in, and stating a thin result honestly
Research	Company filings and calls, peer benchmarks, farm norms, price and inflation data from public sources

Things learnt

- **Size the idea to what the company can manage.** A farm sized on a share of total milk grew too big; a 15-acre pilot is a realistic first step.
- **Judge a backward integration on what it saves the business**, not on a cost-plus price with a capital charge.
- **Learn from peers.** Parag's farm shows how to scale in steps, secure water first and earn the farm's real value through a premium brand.
- **Inflation matters when costs and prices move at different rates.** Feed inflating faster than milk squeezes the saving over time.
- **Herd dynamics are easy to miss.** Replacing a quarter of the herd every year means keeping almost as many young stock as milking cows.
- **Yield and feed decide the case.** A litre of yield or a few percent on feed swings NPV by crores.
- **A 20-litre cow is a lactation average,** not 20 litres every day: yields peak, fall and stop in the dry period.
- **Get the cost of capital right.** Pre-tax savings need a pre-tax rate, and a listed company's WACC needs market-value weights. My first version used the post-tax rate and book weights; correcting both moved NPV from ₹21.4 cr to about −₹1.5 cr and turned "positive" into "roughly break-even".
- **Keep models simple and checkable.** Scenarios and sensitivities that a reader can trace are worth more than clever ones.
- **Working with AI:** AI sped up research, calculations, model building and drafting, while I set the direction, made the decisions, checked the numbers and made sure I understood every step (see the AI and human contribution report).

Project files

- **Captive_Milk_Assessment.docx:** the assessment report presented to management.
- **Milky_Mist_Pilot_Farm_Model.xlsx:** the model (inputs, sales and milk forecast, capital, herd, running costs, savings and NPV, scenarios, sensitivities, cost of capital).
- **Input_Evidence_and_Sources.docx:** the source or basis for every input.
- **AI_and_Human_Contribution_Report.docx:** who did what in the project.